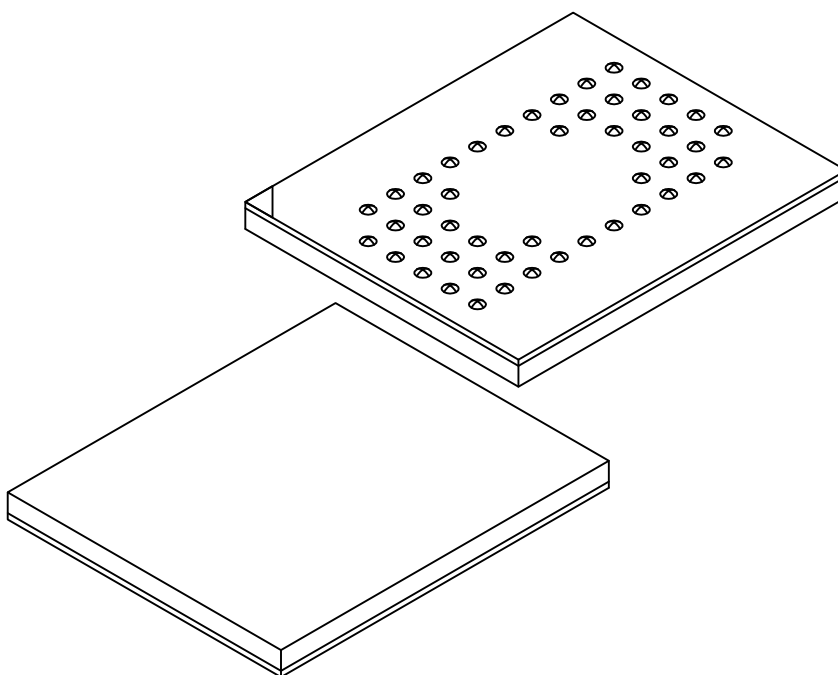


Packaging Diagrams and Parameters

48-Leads Extremely Thin Fine Pitch Land Grid Array (CG) - 5x6 mm Body [XFLGA] SST Legacy Packages C2QE and C2QF

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	48		
Pitch	e	0.50 BSC		
Overall Height	A	-	0.473	0.520
Standoff	A1	0.025	0.04	0.065
Overall Length	D	5.00 BSC		
Overall Terminal Spacing	D1	2.50 BSC		
Overall Width	E	6.00 BSC		
Exposed Pad Width	E2	5.00 BSC		
Terminal Diameter	b	0.24	0.29	0.34

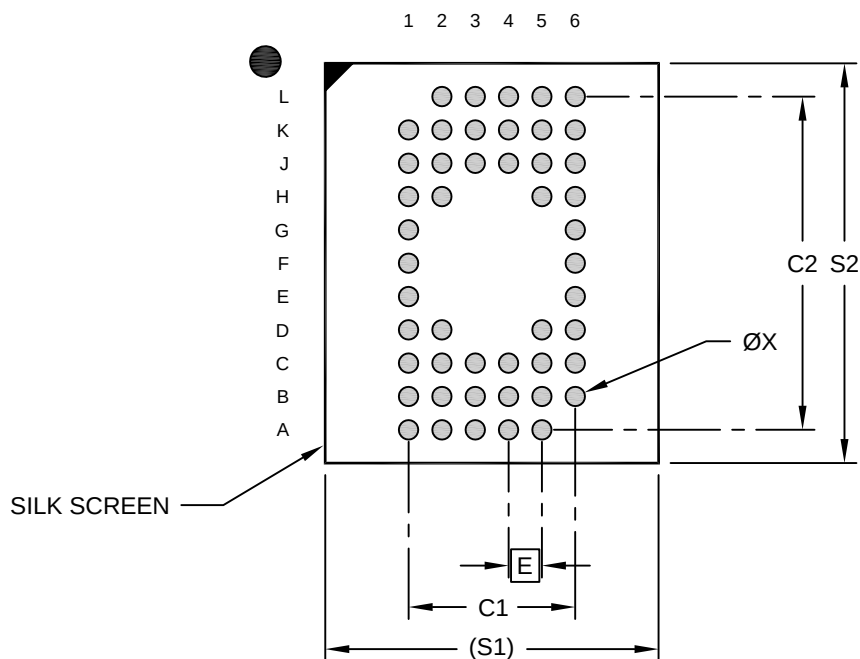
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

48-Leads Extremely Thin Fine Pitch Land Grid Array (CG) - 5x6 mm Body [XFLGA] SST Legacy Packages C2QE and C2QF

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		2.50	
Contact Pad Spacing	C2		5.00	
Contact Pad Diameter (X48)	X			0.29
Package Outline Silk Screen Length	S1	5.00 REF		
Package Outline Silk Screen Width	S2	6.00 REF		

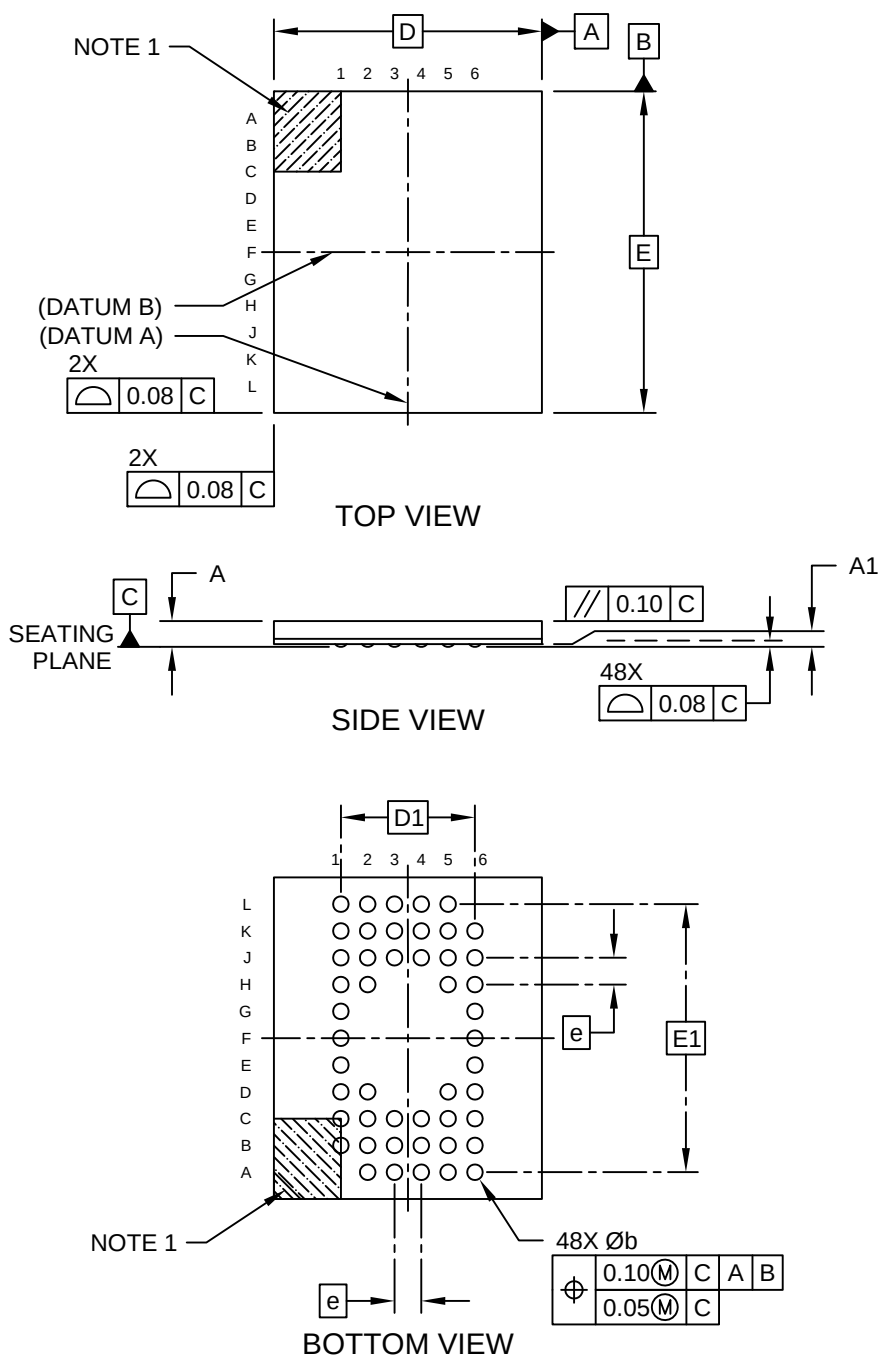
Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Packaging Diagrams and Parameters

48-Leads Extremely Thin Fine Pitch Land Grid Array (CK) - 5x6 mm Body [XFLGA] SST Legacy Packages CBQE and CBQF

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

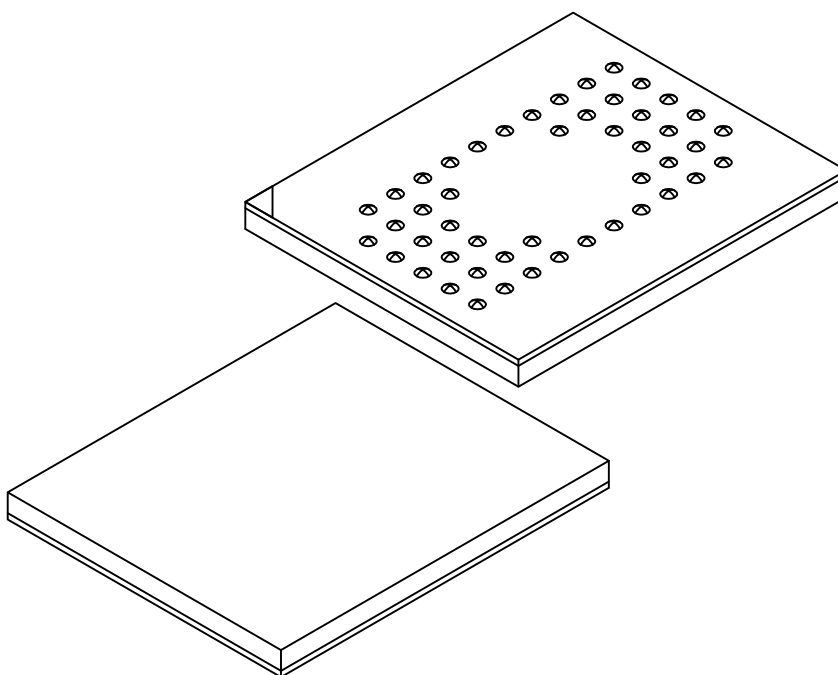


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Packaging Diagrams and Parameters

48-Leads Extremely Thin Fine Pitch Land Grid Array (CK) - 5x6 mm Body [XFLGA] SST Legacy Packages CBQE and CBQF

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		48		
Pitch	e		0.50 BSC		
Overall Height	A	-		0.473	0.520
Standoff	A1		0.025	0.04	0.065
Overall Length	D		5.00 BSC		
Overall Terminal Spacing	D1		2.50 BSC		
Overall Width	E		6.00 BSC		
Exposed Pad Width	E2		5.00 BSC		
Terminal Diameter	b		0.24	0.29	0.34

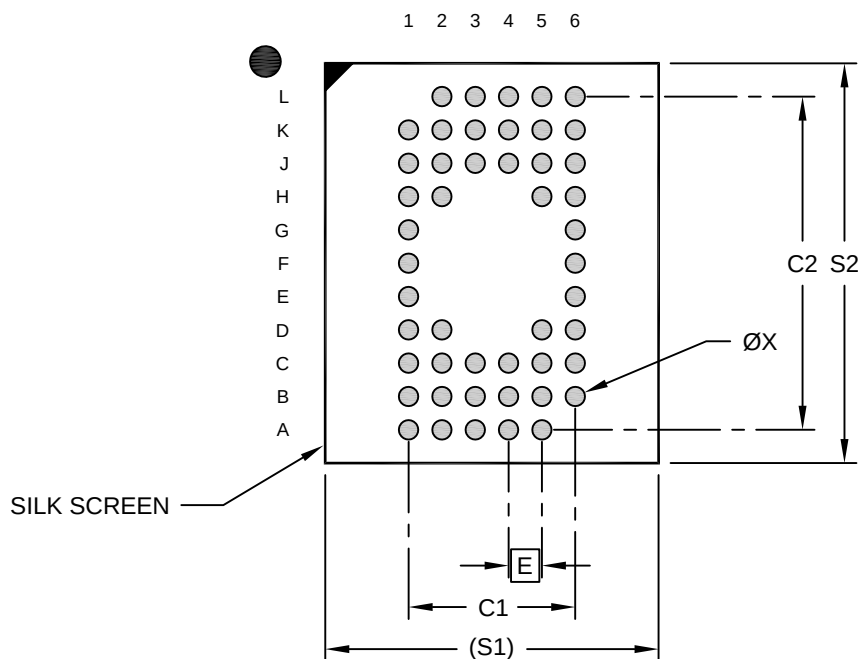
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

48-Leads Extremely Thin Fine Pitch Land Grid Array (CK) - 5x6 mm Body [XFLGA] SST Legacy Packages CBQE and CBQF

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		2.50	
Contact Pad Spacing	C2		5.00	
Contact Pad Diameter (X48)	X			0.29
Package Outline Silk Screen Length	S1	5.00 REF		
Package Outline Silk Screen Width	S2	6.00 REF		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2166-CK Rev A